Lab assignment 1 - Introduction to OpenCV (Open Source Computer Vision Library)

Late Complete

25.8.2025

Attempt 2

Review Feedback
27.8.2025

Attempt 2 Score:
Complete

Add Comment

Anonymous Grading: **no**

Unlimited Attempts Allowed
18.8.2025 to 17.11.2025

Details

Lab assignment 1 (individual)

In this individual lab assignment, every student must complete the following tasks to prepare for the labs in this course.

The code created for the exercises has to be delivered to a code repository of your choice such as Github, Gitlab or SVN, with the following format /IKT213_lastname/assignment_#. Ex. nnoori/IKT213_noori/assignment_1.

SUBMISSION: The submission shall be a link to your git repository along with the outputs from step IV and V below.

I. Setup Anaconda, and an environment with the specified packages:

1. Below are the instructions for setting up an environment using Anaconda: (Anaconda Environment) (https://uia.instructure.com/courses/18389/pages/anaconda-environment)
   1. Download Anaconda3 for your OS https://www.anaconda.com/products/individual (https://www.anaconda.com/products/individual)
   2. Launch the installed **Anaconda Navigator** to create a new environment called **"ikt213"** with the **latest python version** (atleast 3.12+, and the latest version (3.13) should be automatically set when you check the Python box)
   3. Select the environment you've created, and open the dropdown menu to see what packages are **Not installed**; search for the packages "opencv" and "numpy" and click **Apply** at the bottom right.
      1. You will receive a pop-up asking you to apply additional packages, accept these.

II. Setup Git and course directory and add a .gitignore file:

1. Follow the steps in the link if you have not used Git toolkit in any previous courses! https://docs.google.com/document/d/1LCJ-Grf4vh5ND-fjASetrp67TpQ3BWcYkyQiEpA9rSE/edit?rm=embedded (https://docs.google.com/document/d/1LCJ-Grf4vh5ND-

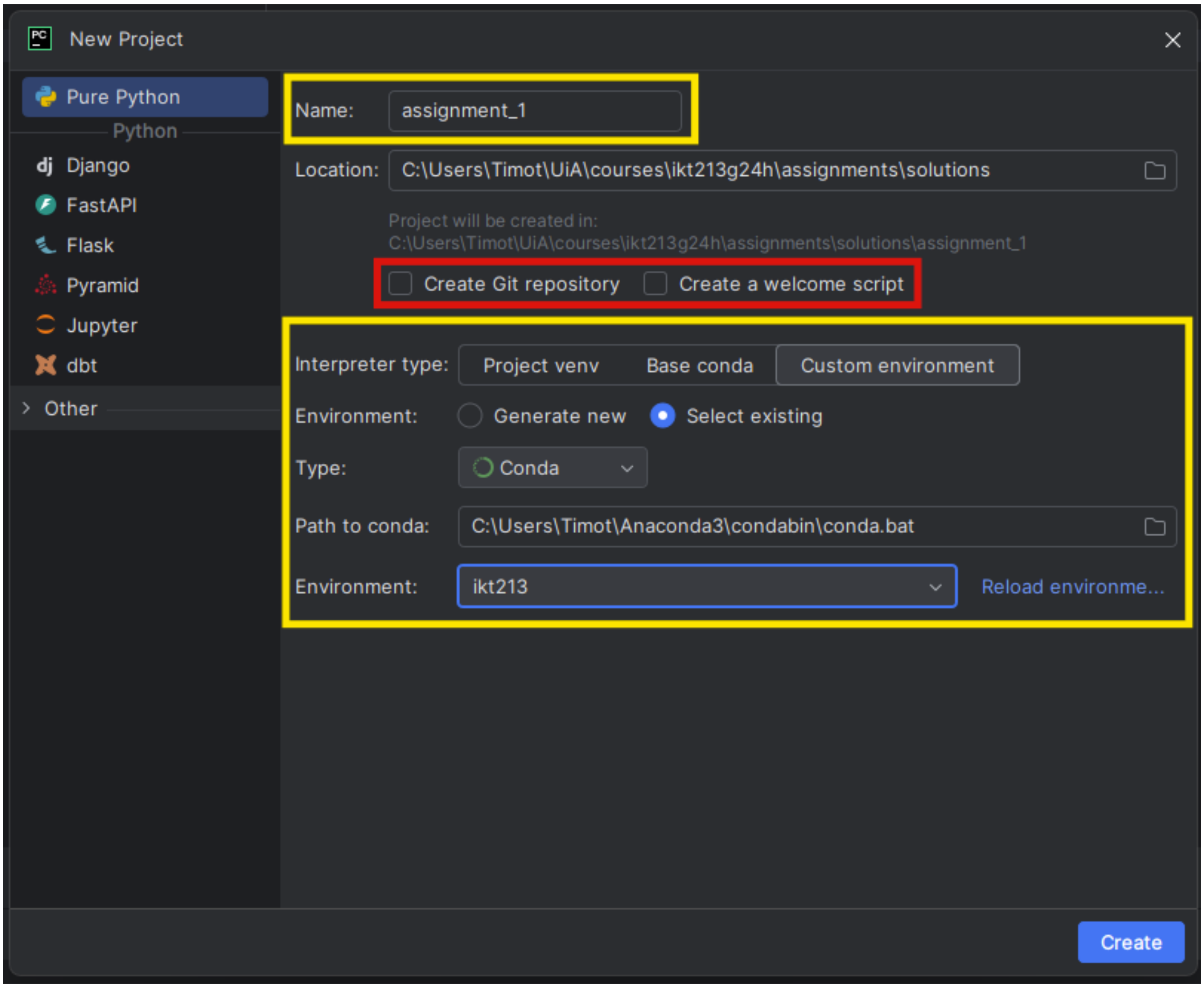
[fjASetrp67TpQ3BWcYkyQiEpA9rSE/edit?rm=embedded)](fjASetrp67TpQ3BWcYkyQiEpA9rSE/edit?rm=embedded)

Now that everything is set up correctly, we can get into the good stuff 🤖!

🟡 III. Assignment directory and project file:

NOTE: PyCharm is just a suggestion, you can use VSCode, Sublime or any other editor of your choice. The steps below would change depending on your editor, feel free to reach out to the TAs if you have questions.

1. Download and install PyCharm [https://www.jetbrains.com/pycharm/ (https://www.jetbrains.com/pycharm/)](https://www.jetbrains.com/pycharm/) (students from UiA will get PyCharm for free by using their UiA account)
2. Create a new project with PyCharm :
   A. You will be using the files that are connected to your git repository.
   B. **NB!!! Uncheck "Create Git repository" and "Create a welcome script"**
   C. For the Interpreter type: Select "Custom environment"
   D. Click "Select existing",
   E. Open the dropdown menu for Type and select "Conda"
   F. Open the dropdown menu for Environment and select "ikt213"
   G. **"Create".**
   (The picture below shows some of the menus we just used.)



(Hot tip 🔥 : run all your functions through a main();)

🟢 **IV. Print image information of the picture "[lena.png (https://uia.instructure.com/courses/18389/files/2994301?wrap=1)](https://uia.instructure.com/courses/18389/files/2994301?wrap=1) (https://uia.instructure.com/courses/18389/files/2994301/download?download_frd=1) " to the console using the OpenCV library (Look at this code [HERE (https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_gui/py_table_of_contents_gui/py_table_of_contents_gui.html#py-table-of-content-gui)](https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_gui/py_table_of_contents_gui/py_table_of_contents_gui.html#py-table-of-content-gui)):**

1. Function name: "**print_image_information**"
2. Function arguments: **image**
   (ex. "**def print_image_information(image):** ")
3. Print the following information using the "**print()**" function from python:
   A. height
   B. width
   C. channels
   D. size (number of values in the cubed array)
   E. data type

Hint: You can use any photo editor software to verify your results.

NOTE: A screenshot of this information has to be uploaded to your repository.

🟢 **V. Web camera information saved into a txt file inside "~/IKT213_lastname/assignment_1/solutions/" (Look at this code [HERE (https://www.geeksforgeeks.org/python-opencv-capture-video-from-camera/)](https://www.geeksforgeeks.org/python-opencv-capture-video-from-camera/)):**

1. File name: **"camera_outputs.txt"**
2. Information saved:
   A. fps
   B. height
   C. width
      (ex: "fps: 69")

NOTE: This text file has to be uploaded to your repository.

Do you need help in the lab or in Discord? Find links here: [Lab Information (https://uia.instructure.com/courses/18389/pages/lab-information)](https://uia.instructure.com/courses/18389/pages/lab-information)